

POWERBOX · DESIGN DECISION RECORD

PowerBox TRL 2 review decisions

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/powerbox		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review and the move to SwapCell interface v0.3

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 12); items 13 to 18 remain proposed, awaiting Amish

■ Context

The TRL 2 review note (docs/REVIEW.md , sessions of 2026-09-24) listed the PowerBox design choices as "Proposed, awaiting Amish", most with a recommendation, and raised two interface issues with SwapCell (waking the pack and pass-through charging). On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." He also approved three cross-cutting additions to the SwapCell interface (a wake method for hosts without CAN, a charge-while-discharging mode and a latch vibration rating for vehicles), a rule that shared SwapCell packs are priced once and excluded from each dependent budget, and a rule that co-design partners are chosen per area later.

This record lists what that instruction decides and what it leaves open because there was no recommendation to accept. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in docs/REVIEW.md (2026-09-24) and PBX-PRC-001 v0.3. They are not repeated here.

■ Decision

Each item in Table 1 is "Decided by Amish, 2026-09-25: go with recommendation" unless the row says otherwise.

Table 1. Decided items.

#	Item	Decision	Where it now lives
1	Pack	Decided by Amish, 2026-09-25: go with recommendation. Removable SwapCell pack; the fixed 12.8 V LiFePO4 pack (about 384 Wh) stays only as a documented fallback	PBX-PRC-001 v0.4, README
2	Bay layout	Decided by Amish, 2026-09-25: go with recommendation. Horizontal side-loading bay with a door on the end	PBX-PRC-001 v0.4, <code>cad/src/model.py</code> , PBX-DWG-001
3	DC input	Decided by Amish, 2026-09-25: go with recommendation. One DC input with MPPT for a solar panel or SunSpoke's panel; a second input only if users need two sources at once	PBX-PRC-001 v0.4, PBX-REQ-001 R3
4	Grid charging	Decided by Amish, 2026-09-25: go with recommendation. Only through a certified external 54.6 V, 5 A charger brick; no AC inlet on the box	PBX-PRC-001 v0.4, PBX-REQ-001 R9
5	AC outlet	Decided by Amish, 2026-09-25: go with recommendation. Include the 300 W AC outlet with auto-off; document a DC-only variant (saves about \$92, no mains voltage)	PBX-PRC-001 v0.4
6	Enclosure	Decided by Amish, 2026-09-25: go with recommendation. Folded aluminium	PBX-PRC-001 v0.4, <code>bom/bom.csv</code>
7	AC region first	Decided by Amish, 2026-09-25: go with recommendation. 230 V 50 Hz with a 30 mA RCD; 120 V 60 Hz is a later variant	PBX-PRC-001 v0.4, PBX-REQ-001 R7, <code>bom/bom.csv</code> lines 9 and 12
8	Budget	Decided by Amish, 2026-09-25: go with recommendation. Keep <code>budget_usd: 450</code> ; it covers the PowerBox parts only. The SwapCell pack is priced once in the SwapCell BOM and excluded here (cross-cutting rule)	<code>project.yaml</code> (unchanged), PBX-REQ-001 R11, <code>bom/bom-notes.md</code>
9	Human-powered charging	Decided by Amish, 2026-09-25: go with recommendation. No human-powered input for PowerBox now; a pedal generator could later be a separate repo using the DC input	PBX-PRC-001 v0.4
10	Pass-through (station mode)	Decided by Amish, 2026-09-25 (cross-cutting approval): SwapCell adds a charge-while-discharging mode. PowerBox cites SwapCell interface v0.3 item C: host type 3 (station), requested mode 4 (charge-discharge), host charge limit in <code>HOST_HEARTBEAT</code>	PBX-PRC-001 v0.4, PBX-CAL-001 section 4
11	Waking the pack	Decided by Amish, 2026-09-25 (cross-cutting approval): SwapCell adds a wake method. PowerBox cites SwapCell interface v0.3 item W: every receiver fits a 10 k Ω coding resistor in its INTERLOCK loop instead of a direct link to ground; no keep-alive cell	PBX-PRC-001 v0.4, PBX-CAL-001 section 4, <code>bom/bom.csv</code> line 5
12	Latch class	Decided by Amish, 2026-09-25 (cross-cutting approval): SwapCell adds latch classes (item V). PowerBox is not a vehicle and uses a class D catch, with the bay door as a second stop; class V1 applies only if PowerBox is ever carried on a vehicle	PBX-PRC-001 v0.4

Items that remain open

These had no recommendation, or depend on users or on SwapCell, so they stay **Proposed, awaiting Amish**.

Table 2. Open items.

#	Item	Status
13	R2 shortfall (1.87 evenings against 2). Options: accept 1.87 evenings and relax R2; discharge to 3.5 % (a 5 % cut-off now gives only 1.97); or revise the reference profile with users (it must fall to about 189 Wh at the loads)	Proposed, awaiting Amish. No recommendation was made at TRL 2
14	First co-design partner (NGO running community charging points, solar home system distributor, or university group)	Proposed, awaiting Amish. Portfolio rule: partners are chosen per area later
15	Theft resistance: is a lockable pack door a priority? The door carries a padlock eye either way	Proposed, awaiting Amish. No recommendation
16	Energy metering on the display for charging points that charge per phone	Proposed, awaiting Amish. No recommendation
17	TRL 3 engineering proposals made in this session: a recessed normally closed wake button in series with the coding resistor, the 30 A buck, and the fuse ratings in PBX-CAL-001	Engineering proposals, awaiting Amish's confirmation
18	Clarification requested from SwapCell: may a pack in legacy discharge (state 5) accept a station heartbeat and move to mode 2 or 4 without opening its output? PowerBox's host is powered from the pack and depends on this	Raised with SwapCell; not changed locally

■ Consequences

- PowerBox now builds to **SwapCell interface v0.3**. The receptacle must carry a $10\text{ k}\Omega \pm 1\%$ INTERLOCK coding resistor; a direct link to SGND would leave the pack output dead.
- The two TRL 2 interface blockers (wake and pass-through) are closed on the SwapCell side. Item 18 is the one remaining interface question.
- The budget stays at \$450 and excludes the pack; the BOM totals \$449 (PBX-CAL-001), so R11 is at risk on indicative prices.
- R2 stays not met until item 13 is decided.